

Lec 8. Attention & Transformers

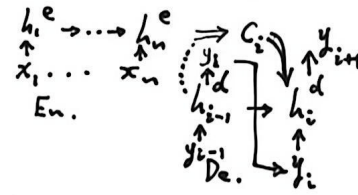
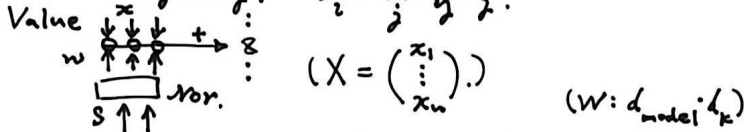
Encoder - Context - Decoder

(LSTM, RNN, TF..) $C = h_n^e$ or <attention mechanism>
 $f(h_1^e, \dots, h_n^e)$

$$c_i = \sum_j \alpha_{ij} h_j^e, \quad \alpha_{ij} = \sigma(s(h_{i-1}^d, h_j^e)) \text{ or } \frac{e^{s(h_{i-1}^d, h_j^e)}}{\sum_j e^{s(h_{i-1}^d, h_j^e)}}.$$

(s is the score func.)

re-weighting: $z_i = \sum_j \alpha_{ij} x_j$



Query $\rightarrow$ $Q = XW^Q, K = XW^K, V = XW^V$

multi-head: $\bigoplus_{n=1}^h$, usually $d_k = d_v = \frac{d_{\text{model}}}{h}$. (or proj. W^o after concat)
 (concat) $(X)(X)$ (d_{model} embedding layer)

TF: 1. input BPE vocab from ind. chars. Train Segmenter
 (subword tokenization) merge two (most freq. adjacent) greedily by order.
 (then modify the corpus) (merges we learned)

2. positional encoding
 Δ necessary? $PE(pos, 2i) = \sin(\frac{pos}{10000 \cdot \frac{2i}{d_{\text{model}}}})$ $+ x \rightarrow x$
 $\dots 2i+1 \dots \cos \dots$

3. input/output Embedding eg. W2V. pretrained or refined during training.

4. multihead attention $Att(Q, K, V) = \sigma(\frac{QK'}{\sqrt{d_k}})V$ (seq-len. seq-len) (seq-len. $\frac{d_{\text{model}}}{h}$)

Opt. mask (before σ) (scale: dot prod. of d_k 's $\propto 1$)
 Δ multi. useful? has variance d_k

5. add & norm Layer Norm (on d_{model}) $LN(x + \text{Dropout}(Att(x)))$
 (sublayer connection) $\gamma \frac{x - \text{mean}(x)}{\text{std}(x)} + \lambda$

6. feed forward (Post LN. or Pre LN)
 $FFN(x) = \text{ReLU}(xW_1 + b_1)W_2 + b_2$ (shifted right)
 (GeLU: $x \cdot \Phi(x)$) (x^+)

7. En.-De. attention Q from decoder input, K, V from encoder (last) output.
 (cross)

8. linear & softmax $W: d_{\text{model}} \cdot |V|$ 9. N each coder:
 (gen. or classify.)

Δ calculate params.

Δ improving attention

$O(n^2) \rightarrow ? O(n)$ performance?

Δ With RNN:
 long range dependency.
 grad vanish & explode.
 parallel computation.

